

SUPPLEMENTARY INFORMATION

T–linear resistivity from magneto-elastic scattering: application to PdCrO₂

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Derivation of the EME coupling

We derive the low-energy effective Kondo lattice model with EME term starting from an Anderson-type model coupled to phonons; the Kondo term was already derived in Ref. [1]. The key new insight is as follows: the effective Kondo-like coupling that is generated between the Pd-electrons and Cr local moments involves an inter-layer hybridization, $t_{cp,ij}$ between sites i and j . If this coupling were to be affected by the fluctuations associated with the corresponding out-of-plane bond, $t_{cp,ij}$ will be renormalized by the displacement, $\varphi_{ij}^{(O)}$. Starting with the microscopic Hamiltonian,

$$H = H_{\text{Pd}} + H_{\text{Cr}} + H_{\text{hyb}} + H_{\text{ph,EME}}, \quad (1)$$

$$H_{\text{Pd}} = \sum_{\mathbf{k}\sigma} (\varepsilon_{\mathbf{k}} - \mu) p_{\mathbf{k}\sigma}^\dagger p_{\mathbf{k}\sigma}, \quad (2)$$

$$H_{\text{Cr}} = - \sum_{ij\sigma} t_{ij}^c c_i^\dagger c_j + \text{h.c.} + U \sum_i \left(n_{i\uparrow}^c - \frac{1}{2} \right) \left(n_{i\downarrow}^c - \frac{1}{2} \right), \quad (3)$$

$$H_{\text{hyb}} = \sum_{\langle ij \rangle \sigma} t_{cp,ij} \left(1 - \eta \varphi_{ij}^{(O)} \right) \left(c_{i\sigma}^\dagger p_{j\sigma} + \text{h.c.} \right) \quad (4)$$

$$H_{\text{ph,EME}} = \sum_{\langle ij \rangle} \frac{\pi_{ij}^{(O)2}}{2M} + \frac{M\tilde{\omega}_0}{2} \varphi_{ij}^{(O)2}, \quad (5)$$

where c , $c^\dagger$ represent the Cr-electron operators, t_{ij}^c is a nearest-neighbor hopping in the Cr layer, $\varepsilon_{\mathbf{k}}$ is the dispersion for the Pd electrons, η determines the scale by which the hybridization is renormalized to leading order in the bond displacement $\varphi_{ij}^{(O)}$, M is the phonon mass, $n_{i\sigma}^c$ is the density operator for Cr electrons, and U is the on site interaction strength. In general, these interactions involve multiple orbitals associated with the Cr crystal field multiplets, and contain the full lattice structure of PdCrO₂. However, here for simplicity, we only retain a single orbital per site in the Cr layers.

We consider the strong-coupling limit, $U \gg t^c, g$ (similarly to Ref. [1]), and derive the low-energy model based on a Schrieffer-Wolff transformation. Retaining terms up to linear-order in the bond displacement, we obtain:

$$H = H_{\text{Pd}} + H_{\text{lm}} + H_{\text{K}} + H_{\text{EME}} + H_{\text{ph,EME}}, \quad (6)$$

$$H_{\text{lm}} = J_H \sum_{\langle ij \rangle} \mathbf{S}_i \cdot \mathbf{S}_j \quad (7)$$

$$H_{\text{K}} = \sum_{\langle ilj \rangle \sigma} \frac{4t_{cp,ij}t_{cp,lj}}{U} \mathbf{S}_l \cdot \left(p_i^\dagger \boldsymbol{\sigma} p_j \right) \quad (8)$$

$$H_{\text{EME}} = - \sum_{\langle ilj \rangle \sigma} \frac{4t_{cp,ij}t_{cp,lj}}{U} \eta \left(\varphi_{il}^{(O)} + \varphi_{jl}^{(O)} \right) \mathbf{S}_l \cdot \left(p_i^\dagger \boldsymbol{\sigma} p_j \right) \quad (9)$$

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The first term above is the usual Heisenberg coupling with antiferromagnetic exchange $J_H = 4(t^c)^2/U$. The second term is a Kondo coupling [1] with the appropriate form-factors coming from the fact that the interaction is inter-layer. The third term is the new EME coupling between local moments in the Cr layer, optical phonons and itinerant Pd-electrons. In what follows, we consider the simplified case where the adjacent Pd layers are stacked in a perfectly aligned arrangement, and the hybridization is independent of the bond direction, i.e. $t_{cp,ij} \rightarrow t_{cp}$. In this limit, we can define the Kondo and EME couplings to be $J_K = 4t_{cp}^2/U$ and $\tilde{\alpha} = -J_K\eta$. We then have the following Hamiltonian in momentum space

$$H = H_{\text{Pd}} + H_{\text{ph}} + H_{\text{lm}} + H_{\text{int}} \quad (10)$$

with the interaction given by

$$H_{\text{int}} = \frac{1}{\sqrt{N}} \sum_{\substack{\mathbf{k}, \mathbf{k}' \\ \alpha\beta}} J_K(\mathbf{k}, \mathbf{k}') p_{\mathbf{k}\alpha}^\dagger [\mathbf{S}_{\mathbf{k}-\mathbf{k}'} \cdot \boldsymbol{\sigma}_{\alpha\beta}] p_{\mathbf{k}'\beta} \quad (11)$$

$$+ \frac{1}{N} \sum_{\mathbf{k}, \mathbf{k}', \mathbf{q}} \sum_{\ell\alpha\beta} \tilde{\alpha}_\ell(\mathbf{k}, \mathbf{k}') \varphi_\ell^{(O)}(\mathbf{k} - \mathbf{k}' - \mathbf{q}) p_{\mathbf{k}\alpha}^\dagger [\mathbf{S}_{\mathbf{q}} \cdot \boldsymbol{\sigma}_{\alpha\beta}] p_{\mathbf{k}'\beta} \quad (12)$$

where the Fourier transformed operators are:

$$\varphi_{\ell+j,j}^{(O)} = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k}\mathbf{R}_j} \varphi_\ell^{(O)}(\mathbf{k}) \quad (13)$$

$$p_{i\alpha}^\dagger = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k}\mathbf{R}_i} p_{\mathbf{k}\alpha}^\dagger \quad (14)$$

$$\mathbf{S}_j = \frac{1}{\sqrt{N}} \sum_{\mathbf{q}} e^{-i\mathbf{q}\mathbf{R}_j} \mathbf{S}_{\mathbf{q}} \quad (15)$$

and the momentum-dependent couplings are

$$J_K(\mathbf{k}, \mathbf{k}') = J_K [2 \cos(k_z a_z)] [2 \cos(k'_z a_z)] \\ \tilde{\alpha}_{+1}(\mathbf{k}, \mathbf{k}') = \tilde{\alpha}_{-1}(\mathbf{k}, \mathbf{k}') = \tilde{\alpha}(\mathbf{k}, \mathbf{k}') = -\eta J_K(\mathbf{k}, \mathbf{k}'),$$

with $\mathbf{R}_i$ being the position of site i and N the number of unit cells.

We have introduced three new parameters: the mass of the out-of-plane mode M (appearing in the EME term), its frequency $\tilde{\omega}_0$ and the EME coupling η . As we detail in the text, these parameters only enter the theory either in the precise combination that corresponds to the dimensionless EME coupling, or in the combination $\beta\tilde{\omega}_0$, leaving effectively two new parameters in the theory. The rest of the parameters are taken from the values provided in [1]. Specifically, we have $U = 4\text{eV}$ and $t_{cp} = 107\text{meV}$ which lead to a Kondo coupling of $J_K = 11.45\text{ meV}$. Additionally, we have $J_H = 6.25\text{meV}$, $t_{nn}^p = 568\text{meV}$, $t_{nnn}^p = 108\text{meV}$, and $\mu = 256\text{meV}$. We fit the difference of the resistivity of PdCrO_2 and PdCoO_2 to obtain the dimensionless EME coupling in the main text and the frequency of this mode.

Transport rate

Within the variational Boltzmann approach [2] the transport lifetime can be estimated by

$$\rho \leq \frac{\frac{V}{2k_B T} \sum_{\mathbf{q}\mathbf{p}} P_{\mathbf{p}\mathbf{q}} (\psi_{\mathbf{q}} - \psi_{\mathbf{p}})^2}{\left| \sum_{\mathbf{q}} e^{\frac{\partial f}{\partial \epsilon_{\mathbf{q}}}} \psi_{\mathbf{q}} \mathbf{v}(\mathbf{q}) \cdot \mathbf{n} \right|^2} \quad (16)$$

with $\psi_{\mathbf{q}}$ a variational ansatz, $P_{\mathbf{p}\mathbf{q}}$ the transition probability from different momentum states, V the volume, $\mathbf{n}$ is a unit vector in the direction of the electric field $\mathbf{E}$, f the Fermi-Dirac distribution, e the electron charge, $\mathbf{v}(\mathbf{q}) = \partial_{\mathbf{q}} \epsilon_{\mathbf{q}}$, and

$\epsilon_{\mathbf{q}}$ the dispersion. Using the ansatz $\psi_{\mathbf{q}} = -e\mathbf{v}_{\mathbf{q}} \cdot \mathbf{E}\tau_{\text{tr}}$ we recover the drude formula with the momentum relaxation time

$$\tau_{\text{tr}}^{-1} = \frac{\frac{1}{k_B T} \sum_{\mathbf{q}\mathbf{p}} P_{\mathbf{p}\mathbf{q}} (\mathbf{v}(\mathbf{q}) \cdot \mathbf{n} - \mathbf{v}(\mathbf{q}) \cdot \mathbf{n})^2}{\sum_{\mathbf{q}} 2 \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{q}}} \right) (\mathbf{v}(\mathbf{q}) \cdot \mathbf{n})^2}. \quad (17)$$

At intermediate- T , where the scattering mechanisms are essentially isotropic, Mathiessen's rule is obeyed such that

$$\tau_{\text{tr}}^{-1} = \tau_{\text{K}}^{-1} + \tau_{\text{EME}}^{-1} + \tau_{\text{el-ph}}^{-1} \quad (18)$$

where each individual scattering rate is determined by the transition probabilities

$$P_{\mathbf{p}\mathbf{q}}^{\text{K}} = k_B T \frac{2\pi}{\hbar} \left(-\frac{\partial f_{\mathbf{p}}}{\partial \epsilon_{\mathbf{p}}} \right) |J_{\text{K}}(\mathbf{p}, \mathbf{q})|^2 \chi''(\mathbf{p} - \mathbf{q}, \epsilon_{\mathbf{p}} - \epsilon_{\mathbf{q}}) (n_B(\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{q}}) + n_F(-\epsilon_{\mathbf{q}})), \quad (19)$$

$$P_{\mathbf{k}\mathbf{p}}^{\text{EME}} = k_B T \frac{2\pi}{\hbar} \left(-\frac{\partial f}{\partial \epsilon_{\mathbf{p}}} \right) |\tilde{\alpha}(\mathbf{p}, \mathbf{q})|^2 \sum_{\mathbf{k}} \int \frac{d\nu}{\pi} \chi''(\mathbf{k} + \mathbf{p} - \mathbf{q}, \epsilon_{\mathbf{p}} - \nu) \mathcal{B}(\mathbf{k}, \nu - \epsilon_{\mathbf{q}}) \times \\ [n_B(\epsilon_{\mathbf{p}} - \nu) + n_F(-\nu)] [n_B(\nu - \epsilon_{\mathbf{q}}) + n_F(-\epsilon_{\mathbf{q}})], \quad (20)$$

$$P_{\mathbf{p}\mathbf{q}}^{\text{el-ph}} = k_B T \frac{2\pi}{\hbar} \left(-\frac{\partial f_{\mathbf{p}}}{\partial \epsilon_{\mathbf{p}}} \right) \sum_{\lambda} |\alpha_{\lambda}(\mathbf{p}, \mathbf{q})|^2 \mathcal{B}_{\lambda}(\mathbf{p} - \mathbf{q}, \epsilon_{\mathbf{p}} - \epsilon_{\mathbf{q}}) (n_B(\epsilon_{\mathbf{p}} - \epsilon_{\mathbf{q}}) + f(-\epsilon_{\mathbf{q}})). \quad (21)$$

Here $\chi''(\mathbf{p}, \omega)$ is the spin susceptibility and $\mathcal{B}_{\lambda}(\mathbf{k}, \nu)$ is the phonon spectral function for mode $\lambda = o, a$ which can be either acoustic or optical. We also reintroduced the momentum dependence in all the couplings. In the case where the magnetic and itinerant electron layers are perfectly aligned we have

$$J_{\text{K}}(\mathbf{p}, \mathbf{q}) = J_{\text{K}} [2 \cos(p_z a_z)] [2 \cos(q_z a_z)], \quad (22)$$

$$\tilde{\alpha}(\mathbf{p}, \mathbf{q}) = \tilde{\alpha} [2 \cos(p_z a_z)] [2 \cos(q_z a_z)], \quad (23)$$

$$\alpha_{\text{o}}(\mathbf{p}, \mathbf{q}) = \alpha_{\text{o}}, \quad (24)$$

and

$$\alpha_{\text{a}}(\mathbf{p}, \mathbf{q}) = \alpha_{\text{a}} |\mathbf{p} - \mathbf{q}|. \quad (25)$$

Here the coupling constants are as defined in the main text. Additionally, we make the local approximation for the spins [3]:

$$\chi''(\nu) = S(S+1) \beta \nu \frac{\theta(2\pi J_{\text{H}} - |\nu|)}{2J_{\text{H}}} \quad (26)$$

with the sum rule constraint

$$\int \frac{\chi''(\nu)}{\beta \nu} d\nu = 2\pi S(S+1). \quad (27)$$

The spectral function for the free phonons is given by

$$\mathcal{B}_{\lambda}(\mathbf{k}, \nu) = \frac{\pi \hbar}{2M\omega_{\mathbf{k},\lambda}} (\delta(\nu - \omega_{\mathbf{k},\lambda}) - \delta(\nu + \omega_{\mathbf{k},\lambda})) \quad (28)$$

where for optical phonons we have $\omega_{\mathbf{k},o} = \omega_0$ while for acoustic phonons $\omega_{\mathbf{k},a} = ck$. The integrals above can then be evaluated analytically if we take a quasi-2D circular fermi surface. This leads to the total scattering rate:

$$\tau_{\text{tr}}^{-1} = 2\pi \frac{\lambda_{\text{tr,EP,o}}}{\hbar \beta} \mathcal{I}_1 \left(\frac{\beta \hbar \omega_0}{2} \right) + 2\pi \frac{\lambda_{\text{tr,EP,a}}}{\hbar \beta} \mathcal{I}_2 \left(\frac{\beta \hbar \omega_{BG}}{2} \right) \\ + 2\pi \frac{\lambda_{\text{K}} J_{\text{K}}}{\hbar} \mathcal{I}_3(2\pi \beta J_{\text{H}}) + 2\pi \frac{\lambda_{\text{tr,EME}}}{\hbar \beta} \mathcal{I}_4(2\pi \beta J_{\text{H}}, \beta \hbar \tilde{\omega}_0) \quad (29)$$

Here we define

$$\mathcal{I}_1(x) = [x \operatorname{csch}(x)]^2, \quad (30)$$

and

$$\mathcal{I}_2(x) = \frac{4}{\pi} \frac{1}{x^3} \int_0^x dy \frac{y^4}{\sqrt{1 - (\frac{y}{x})^2}} [\operatorname{csch}(y)]^2, \quad (31)$$

corresponding to the case of acoustic phonons interacting with a circular Fermi surface in two dimensions. In addition,

$$\mathcal{I}_3(y) = -\frac{4\operatorname{Li}_2(e^{-y})}{y} - \frac{2y}{e^y - 1} - 4y + \frac{2\pi^2}{3y} + 4\log(e^y - 1), \quad (32)$$

corresponds to the pure Kondo scattering at high temperatures, where the logarithmic integrals result from integrating the Fermi function. And finally

$$\mathcal{I}_4(y, z) = \frac{z^2}{\cosh(z) - 1} \mathcal{J}(y, z) \quad (33)$$

is the result of the integrals involving the EME coupling, with the additional definition:

$$\begin{aligned} \mathcal{J}(y, z) = & \frac{-2z\operatorname{Li}_2(e^{-y}) + (2y + z)\operatorname{Li}_2(e^{-y-z}) + (z - 2y)\operatorname{Li}_2(e^{z-y}) + 2\operatorname{Li}_3(e^{-y-z}) - 2\operatorname{Li}_3(e^{z-y})}{2yz} \\ & + \frac{z^2}{12y} - \frac{1}{2} \log((e^y - e^z)(e^{y+z} - 1)) + \frac{y \log(\sinh(\frac{y-z}{2}) \operatorname{csch}(\frac{y+z}{2}))}{2z} + \frac{y}{2} + \frac{\pi^2}{3y} + \log(e^y - 1) + \frac{z}{2}. \end{aligned} \quad (34)$$

These functions have the following high temperature limits which can be obtained from the asymptotic expansions for the polylogarithm

$$\lim_{T \rightarrow \infty} \mathcal{I}_1\left(\frac{\beta \hbar \omega_0}{2}\right) = \lim_{T \rightarrow \infty} \mathcal{I}_2\left(\frac{\beta \hbar \omega_{BG}}{2}\right) = 1 \quad (35)$$

and

$$\lim_{T \rightarrow \infty} \mathcal{I}_3(2\pi\beta J_H) = \lim_{T \rightarrow \infty} \mathcal{I}_4(2\pi\beta J_H, \beta \hbar \tilde{\omega}_0) = 2. \quad (36)$$

In the main text, we use the effective mass of PdCoO₂ extracted from quantum oscillations given in Ref. [4], and the density corresponding to half filling to extract the transport rate from experiments following the procedure in Ref. [5] to calculate the ratio n/m for both PdCoO₂ and PdCrO₂. Then we fit the resulting scattering rate to the first two terms which correspond to electron-phonon scattering in τ_{tr}^{-1} . With this procedure, we find the parameters $\lambda_{el-ph,o} = 0.024, \hbar\omega_0 = 120\text{meV}, \lambda_{el-ph,a} = 0.043$, and $\hbar\omega_{BG} = 29\text{meV}$, which are in close agreement with the reported values in [6, 7]. The difference between our treatment and that in Refs. [6, 7] is that we consider a 2D fermi surface instead of a 3D Fermi surface. At intermediate temperatures where our theory is valid we do not see significant differences in the fit quality between the model considered here and the ones in [6, 7].

We then carry out the same procedure for PdCrO₂ and subtract the resulting scattering rate from the scattering rate extracted from PdCoO₂. This would isolate the last two terms that correspond to Kondo and EME scattering in τ_{tr}^{-1} . By fitting this excess scattering rate we find the EME parameters $\lambda_{EME} = 0.025$ and $\hbar\tilde{\omega}_0 = 45\text{meV}$. Since we use the high temperature approximation for χ'' we do not fit the low temperature regime of $\Delta_\tau = \tau_{tr,Cr}^{-1} - \tau_{tr,Co}^{-1}$ but only carry the fit at temperatures above 80K ($J_H \sim 72K$). Note that the constant offset between the PdCoO₂ and the PdCrO₂ resistivities is not fitted. This excess scattering rate comes purely from the Kondo scattering with the dimensionless coupling $\lambda_K = 4S(S+1)J_K\nu_0$ and a saturation scale set by J_H , where the value of these couplings is listed above.

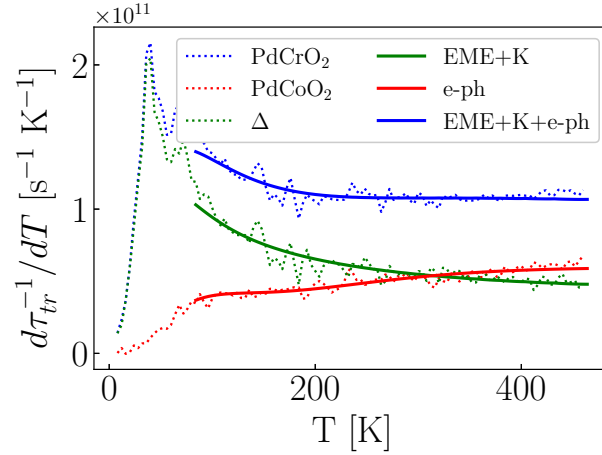
In the above, one can trace the various features of ρ for PdCrO₂ to each of the terms in Eqn. 29. The Kondo term increases sharply and completely saturates at a temperature scale set by $2\pi J_H$. This approach to saturation would be responsible for the sublinear approach to the T -linear regime. This sublinear piece dominates at $T \sim J_H$ over the contributions from the phonon and the magnetoelastic scattering rates. Eventually, both of these contributions catch up with the Kondo scattering rate and the resistivity becomes T -linear which can be seen from the high temperature

limits of $\mathcal{I}_4$ and $\mathcal{I}_3$ and the expression for τ_{tr}^{-1} . Taking the high temperature limit, we can determine the coefficient of the linear in T scattering rate. Using the asymptotic expressions above and the values of the fitted dimensionless couplings we get

$$\lim_{T \rightarrow \infty} \frac{d}{dT} \tau_{tr}^{-1} = \left(\frac{k_B}{\hbar} \right) 2\pi (\lambda_{el-ph,o} + \lambda_{el-ph,a} + 2\lambda_{EME}) \quad (37)$$

$$\sim 0.75 \left(\frac{k_B}{\hbar} \right) \quad (38)$$

which is close to the Planckian slope $k_B/\hbar$.

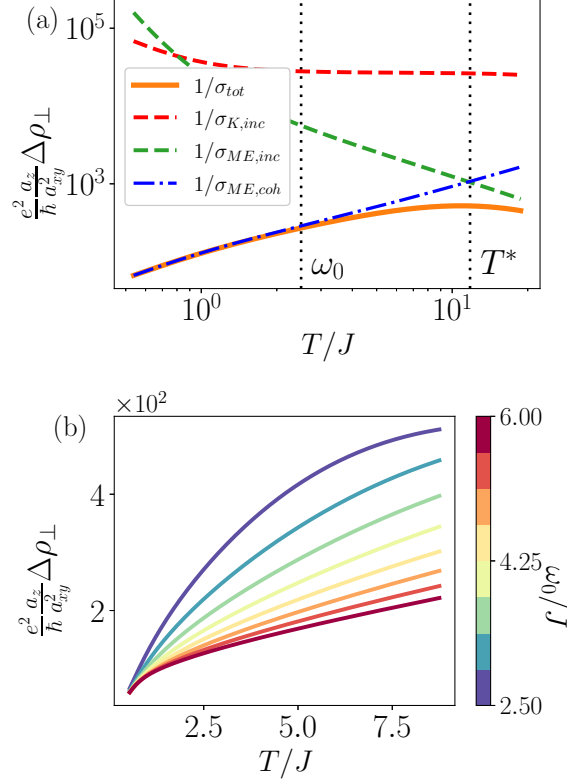


Supplemental Figure S1. Rate of change of the scattering rates from fit and experiments. In the case where the resistivity is completely saturated we expect this quantity to be constant. However, neither the electron-phonon contribution, nor the Kondo and EME contributions are linear. Nevertheless, the sum of the scattering rates is almost linear, as evidenced by the constant rate of change.

Finally, we note that neither τ_{el-ph}^{-1} nor τ_K^{-1} are in the fully saturated regime for the values of the fit parameters obtained from the above procedure (see Fig. S1). In contrast, the excess resistivity from the EME term is generated by a mode that saturates at relatively low temperatures (since the out-of-plane vibration in PdCrO₂ turns out to be softer than the in-plane one). At the same time, the addition of the pure Kondo scattering gives an additional T -sublinear contribution to the resistivity. In total, the T -superlinear scattering rate from electron-phonon scattering and the T -sublinear contribution from the Kondo term, together with the T -linear EME contribution yield an approximately T -linear resistivity for PdCrO₂.

Out-of-plane resistivity

The starting point of our discussion is a slightly modified version of the model introduced in the discussion of in-plane transport. Here, we assume that in addition to the regular in-plane hopping in the Pd layer, we have a small but non negligible hopping t_{pp} between adjacent Pd layers. The value of this hopping parameter is estimated to be $\sim 0.042t_{nn}^p$ by taking as a reference the dispersion of PdCoO₂ as measured by quantum oscillations [6]. Once this hopping has been introduced, the current operator in the z direction can be readily calculated. This operator has



Supplemental Figure S2. c-axis transport: (a) different contributions to the c-axis resistivity for a soft phonon frequency $\omega_0 = 2.5J_H$. Coherent transport dominates over a broad temperature regime but deviates from T linearity at a temperature T_* set by the electron velocity and the magnetoelastic coupling (see main text). The pure incoherent Kondo scattering is always shorted by other conduction channels. (b) Dependence of the c-axis resistivity on variations of the phonon frequency. As ω_0 becomes softer the incoherent conductivity is enhanced and the coherent scattering increases making a saturation regime accessible at high temperatures.

three distinct contributions (ignoring for simplicity the phonon-assisted, but local-moment independent, term):

$$\mathbf{J}^0 = -i \sum_{ij\sigma} t_{ij}^p (\mathbf{r}_i - \mathbf{r}_j) p_{i\sigma}^\dagger p_{j\sigma} \quad (39)$$

$$\mathbf{J}^1 = -i \frac{J_K}{2} \sum_{ilj\beta\alpha} (\mathbf{r}_i - \mathbf{r}_j) p_{i\alpha}^\dagger (\mathbf{S}_l \cdot \boldsymbol{\sigma}_{\alpha\beta}) p_{j\beta} \quad (40)$$

$$\begin{aligned} \mathbf{J}^2 = i \frac{\eta J_K}{2} \sum_{ilj\beta\alpha} & \left(\varphi_{il}^{(O)} + \varphi_{jl}^{(O)} \right) \\ & \times (\mathbf{r}_i - \mathbf{r}_j) p_{i\alpha}^\dagger (\mathbf{S}_l \cdot \boldsymbol{\sigma}_{\alpha\beta}) p_{j\beta} \end{aligned} \quad (41)$$

where $\mathbf{J}^0$ is the usual current operator that has a z component induced by t_{pp} , $\mathbf{J}^1$ is an assisted hopping current mediated by a local moment $\mathbf{S}_l$ at site $\mathbf{r}_l$ in the Cr layer, and $\mathbf{J}^2$ is a phonon and local moment assisted hopping, with $\varphi_{il}^{(O)}$ being the bond stretching displacement between sites at $\mathbf{r}_i$ in a Pd layer and $\mathbf{r}_l$ in a Cr layer. For the in-plane transport, these conduction channels are usually shorted by the coherent current $\mathbf{J}^0$ due to the large in-plane Fermi velocity, which is why they were absent from our previous discussions. However, these terms are relevant in the discussion of c-axis resistivity, since the hopping integral t_{pp} is now of similar magnitude as the Kondo and EME couplings which set the overall scale of the incoherent currents.

To calculate the conductivity we compute the current-current correlation for the total current $\mathbf{J} = \sum_i \mathbf{J}^i$. Since we work perturbatively in the Kondo and EME couplings, as well as in the out-of-plane hopping t_{pp} , the cross correlations

between different current operators are subleading and we obtain

$$\sigma^c = \sigma_{\text{inc}} + \sigma_{\text{coh}} \quad (42)$$

$$\sigma_{\text{inc}}^c = \lim_{\omega \rightarrow 0} \lim_{\mathbf{q} \rightarrow 0} \sum_{i=1,2} \frac{-1}{\beta V} \langle J_z^i(\tilde{\mathbf{q}}) J_z^i(-\tilde{\mathbf{q}}) \rangle \quad (43)$$

$$\sigma_{\text{coh}}^c = 4e^2 t_{pp}^2 a_z^2 \nu_0 \langle \tau_{\text{tr}} \sin^2 k_z a_z \rangle \quad (44)$$

where $\langle \cdot \rangle$ denotes the momentum average over the Fermi surface, and ν_0 is the density of states at the Fermi level. The behavior of each individual contribution to the conductivity is shown in Fig. S2(a). Here we see that the incoherent contribution from the Kondo current $\mathbf{J}^1$ to the resistivity decreases with increasing T and then saturates at a scale set by J_H . In contrast the contribution of the EME current $\mathbf{J}^2$ to the resistivity continues to decrease with temperature with $\rho_{\text{inc, EME}} \sim T^{-1}$. At the same time the coherent contribution rises with temperature $\rho_{\text{coh}} \sim T$. We then obtain the total resistivity from parallel addition of each individual conduction channel. Interestingly, we note that there is a crossover temperature scale T_* at which the incoherent and coherent resistivities coincide. At this scale, we expect resistivity saturation. At asymptotically high temperatures, there is an eventual change of trend where the resistivity decreases as a function of temperature as a consequence of incoherent processes.

To estimate the scale at which this crossover occurs, we simply equate the high temperature limits of σ_{inc}^c and σ_{coh}^c . Since the incoherent conductivity is dominated by the EME current correlations, we can approximate at high temperatures:

$$\sigma_{\text{inc}}^c \sim \left(\frac{a_z}{a_{xy}^2} \frac{e^2}{\hbar} \right) \lambda_{\text{EME}} \nu_0 T \quad (45)$$

In the same regime, the coherent contribution to the conductivity can be well approximated by:

$$\sigma_{\text{coh}}^c \sim \left(\frac{a_z}{a_{xy}^2} \frac{e^2}{\hbar} \right) \frac{t_{pp}^2 \nu_0}{(\lambda_{\text{EME}} + \lambda_{\text{el-ph}}) k_B T}. \quad (46)$$

Equating these two conductivities and estimating $v_z \sim t_{pp} a_z / \hbar$ we get the crossover temperature

$$T_* \sim \frac{v_z G_z}{\sqrt{\lambda_{\text{EME}}(\lambda_{\text{EME}} + \lambda_{\text{el-ph}})}}, \quad (47)$$

with $G_z = 2\pi\hbar/a_z$. Note that the dependence on the density of states is now hidden in λ_{EME} , which also depends on the frequency of the mode that mediates the EME interaction.

We now analyze the effect of pressure on the c -axis resistivity. Specifically, applying compressive strain along the c direction introduces changes in the lattice constants, the hopping integrals, the Kondo coupling, and the phonon frequency. The expectation is that the phonon frequencies and lattice constants change algebraically while J_K and t_{pp} change exponentially, so we focus on the latter in what follows. Additionally, we have $J_K \propto t_{cp}^2/U$, with t_{cp} setting the scale of the hybridization between Cr and Pd layers. Consider the case where t_{cp}^2 has an exponential dependence on a_z similar to that of t_{pp} . This could happen, for instance, if the inter-layer hopping is controlled by virtual tunneling via the Cr layers, then t_{pp} is also proportional to t_{cp}^2/U . As a consequence, J_K increases at the same rate as t_{pp} when pressure is applied along the c -axis. This entails that upon applying pressure, σ_{inc}^c grows since the overall scale of the incoherent current increased. At the same time, σ_{coh}^c increases since the increase in the c -axis velocity is matched by the increase of the scattering rate with decreasing a_z for the Kondo and EME contributions, but not for the el-ph contribution.

These changes to the resistivity also have an effect on the crossover scale T_* . With the scaling of the different hopping parameters, we find that λ_{EME} scales with c -axis pressure in the same way as v_z^2 . Following Eqn. 47, the crossover temperature T_* is expected to decrease with increasing pressure. As such, deviations from T -linearity in the resistivity are expected to occur at lower temperatures when pressure is applied.

Incoherent contribution to the c -axis resistivity

To evaluate the incoherent contribution to the conductivity we start by Fourier transforming the current operators that are associated with the Kondo and EME assisted hoppings between Pd layers. These current operators are given

by

$$J_z^1(\omega_n) = \frac{1}{\sqrt{N\beta}} \sum_{\substack{\mathbf{k}\mathbf{k}'\alpha\beta \\ i\nu_l, i\nu_n}} f(\mathbf{k}, \mathbf{k}') p_{\mathbf{k}\alpha}^\dagger(i\nu_l) [\mathbf{S}_{\mathbf{k}-\mathbf{k}'}(i\nu_l - i\nu_n) \cdot \boldsymbol{\sigma}_{\alpha\beta}] p_{\mathbf{k}'\beta}(i\nu_n + i\omega_n) \quad (48)$$

$$J_z^2(\omega_n) = \frac{1}{N\beta} \sum_{\substack{\mathbf{k}\mathbf{k}'\mathbf{q}\alpha\beta\ell \\ i\nu_l, i\mu_m, i\nu_n}} \Lambda_\ell(\mathbf{k}, \mathbf{k}') \varphi_{\ell, \mathbf{k}-\mathbf{k}'-\mathbf{q}}^{(O)}(i\nu_l - i\mu_m - i\nu_n) p_{\mathbf{k}\alpha}^\dagger(i\nu_l) [\mathbf{S}_{\mathbf{q}}(i\mu_m) \cdot \boldsymbol{\sigma}_{\alpha\beta}] p_{\mathbf{k}'\beta}(i\nu_n + i\omega_n) \quad (49)$$

where the bare vertices are given by

$$f(\mathbf{k}, \mathbf{k}') = J_K \left(\frac{2a_z}{\hbar} \right) \sin(k_z a_z + k'_z a_z), \quad (50)$$

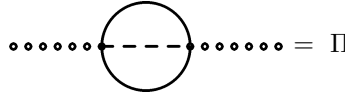
and

$$\Lambda_+(\mathbf{k}, \mathbf{k}') = \Lambda_-(\mathbf{k}, \mathbf{k}') = -\eta f(\mathbf{k}, \mathbf{k}') \quad (51)$$

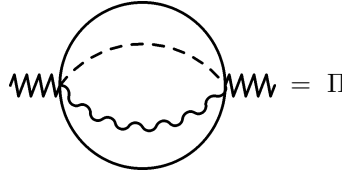
we additionally parametrized the magnetoelastic interaction as $\tilde{\alpha} = -J_k \eta$. Note that we have two phonon operators per local moment since there are two bond stretching modes, one that connects to the upper and one that connects to the lower Pd layers. With these definitions, we proceed to calculate the current-current correlation functions

$$\Pi_{\mathbf{k}}^{zz}(i\omega_n) = \frac{-1}{\beta V} \sum_i \langle J_z^i(\tilde{q}) J_z^i(-\tilde{q}) \rangle = \Pi_1 + \Pi_2 \quad (52)$$

There are two diagrams that contribute to leading order, these are respectively:



$$\text{Diagram 1} = \Pi_1 \quad (53)$$



$$\text{Diagram 2} = \Pi_2 \quad (54)$$

where the dotted line corresponds to the $\mathbf{J}^1$ current, the dashed line denotes the spin susceptibility, the zig-zag line is a $\mathbf{J}^2$ current, and the wiggly line is a phonon propagator. Note that there are no other phonon vertices since the interaction is diagonal in the phonon mode. If we allow for anharmonicities, further corrections to the Π_2 correlation function will contribute.

The first diagram above amounts to the following expression:

$$\Pi_{\mathbf{k},1}^{zz}(i\omega_n) = \frac{1}{\beta V} \frac{1}{\beta N} \sum_{\mathbf{qp}} \sum_{i\nu_n, iz_n} |f(\mathbf{p} + \mathbf{q}, \mathbf{k} + \mathbf{q})|^2 S(\mathbf{p}, iz_n) \mathcal{G}(\mathbf{p} + \mathbf{q}, i\nu_n + iz_n) \mathcal{G}(\mathbf{k} + \mathbf{q}, i\omega_n + i\nu_n). \quad (55)$$

We expect that in the high temperature limit vertex corrections can be safely ignored in the perturbative regime when quasi-elastic collisions dominate and there is no momentum structure to the scattering potentials (since the spins are completely disordered). In this case, we proceed by performing the bosonic and fermionic Matsubara summations:

$$\begin{aligned} \Pi_{\mathbf{k},1}^{zz}(i\omega_n) &= \frac{1}{V} \frac{1}{N} \sum_{\mathbf{q}\mathbf{p}} |f(\mathbf{p} + \mathbf{q}, \mathbf{k} + \mathbf{q})|^2 \int \frac{d\epsilon}{\pi} \int \frac{d\nu}{\pi} \int \frac{dz}{\pi} \chi''(\mathbf{p}, z) A(\mathbf{p} + \mathbf{q}, \nu) A(\mathbf{k} + \mathbf{q}, \epsilon) \\ &\quad \times \left(\frac{n_F(\epsilon) - n_F(\nu - z)}{i\omega_n - \epsilon + \nu - z} \right) (n_B(z) + n_F(\nu)). \end{aligned} \quad (56)$$

We see that overall the current correlation function is a convolution of electron spectral functions with the spin susceptibility. Once we perform analytic continuation and take the zero frequency limit, we get

$$\sigma_{11}^c = \frac{\hbar e^2}{VN} \sum_{\mathbf{q}\mathbf{p}} |f(p_z, q_z)|^2 \int \frac{d\epsilon}{\pi} \left[\int \frac{dz}{\pi} \chi''(\epsilon - z) A(\mathbf{p}, z) (n_B(\epsilon - z) + n_F(-z)) \right] A(\mathbf{q}, \epsilon) \left(-\frac{\partial n_F(\epsilon)}{\partial \epsilon} \right), \quad (57)$$

where we already took the $\mathbf{k} \rightarrow 0$ limit and we used the local approximation on χ'' to simplify the momentum sums. Since the electron spectral functions satisfy $A(\mathbf{p}, \omega) \approx A((p_x, p_y), \omega)$ we can pull the momentum sums over $q_{x,y}$ and $p_{x,y}$ past the bare vertex, and to leading order we have

$$\sigma_{11}^c = \frac{2a_z}{a_{pl}^2} \hbar e^2 \int \frac{dp_z}{2\pi} \frac{dq_z}{2\pi} |f(p_z, q_z)|^2 \int \frac{d\epsilon}{\pi} \left[\int \frac{dz}{\pi} \chi''(\epsilon - z) \rho(z) (n_B(\epsilon - z) + n_F(-z)) \right] \rho(\epsilon) \left(-\frac{\partial n_F(\epsilon)}{\partial \epsilon} \right) \quad (58)$$

using

$$\sigma_{11}^c = \left(\frac{2a_z}{a_{pl}^2} \frac{e^2}{\hbar} \right) J_K^2 \int \frac{d\epsilon}{\pi} \rho(\epsilon) \left(-\frac{\partial n_F(\epsilon)}{\partial \epsilon} \right) \int \frac{dz}{\pi} \rho(\epsilon - z) \chi''(z) (n_B(z) + n_F(z - \epsilon)).$$

To evaluate this integral we use the density of states evaluated from the dispersion in the low energy theory of [1].

Now we focus on the EME current correlation functions

$$\Pi_{\mathbf{q},2}^{zz}(i\omega_n) = -\frac{1}{\beta^3 V N^2} \sum_{\substack{\mathbf{k}'\mathbf{k}\mathbf{p} \\ ik'_n ik_n ip_n \\ \ell}} |\Lambda_\ell(\mathbf{k}, \mathbf{k}' + \mathbf{q})|^2 D(\mathbf{k} - \mathbf{k}' - \mathbf{p}, ik_n - ip_n - ik'_n) S(\mathbf{p}, ip_n) \mathcal{G}(\mathbf{k}, ik_n) \mathcal{G}(\mathbf{k}' + \mathbf{q}, ik'_n + iq_n).$$

Note that D is independent of ℓ assuming that the frequencies of the two EME modes per unit cell are roughly equal. This is a consequence of the fact that the mass and the frequency of the two optical modes for the outgoing bonds from a local moment in the Cr layer are the same. Once again we can perform all three Matsubara sums, we then perform analytic continuation and take the appropriate DC limit, which gives the following expression for the c-axis conductivity:

$$\begin{aligned} \sigma_{22}^c &= -\frac{e^2}{VN} \sum_{\mathbf{q}\mathbf{k}\ell} \int \frac{d\epsilon}{\pi} \frac{d\mu}{\pi} \frac{dz}{\pi} |\Lambda_\ell(\mathbf{q}, \mathbf{k})|^2 B(z) \chi''(\mu) A(\mathbf{q}, \epsilon + z + \mu) A(\mathbf{k}, \epsilon) \times \\ &\quad [n_B(-z) - n_B(\mu)] [n_B(z + \mu) + n_F(\epsilon + z + \mu)] \left(-\frac{\partial n_F(\epsilon)}{\partial \epsilon} \right). \end{aligned} \quad (59)$$

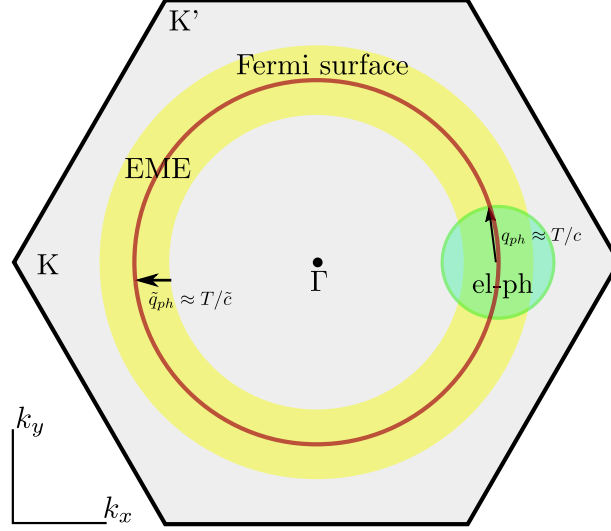
Note that we have assumed to be a dispersion-less Einstein mode, which simplifies the momentum integrations. Once again, in the quasi 2D limit, we can perform the momentum sums first, which approximately converts each factor of the spectral function into a density of states. Finally, we use the phonon spectral function to evaluate one of the frequency integrals, which gives the result:

$$\begin{aligned} \sigma_{22}^c &= \left(\frac{2a_z}{a_{pl}^2} \frac{e^2}{\hbar} \right) \left(\frac{\hbar \eta^2}{M\omega_0} \right) [2J_K]^2 \text{csch}\left(\frac{\beta\omega_0}{2}\right) \int \frac{d\epsilon}{\pi} \frac{d\mu}{\pi} \rho(\epsilon) \cosh\left(\frac{\beta\epsilon}{2}\right) \left(-\frac{\partial n_F(\epsilon)}{\partial \epsilon} \right) \times \\ &\quad S(\mu) \left[\rho(\epsilon + \omega_0 + \mu) \text{sech}\left(\beta \frac{\epsilon + \mu + \omega_0}{2}\right) + \rho(\epsilon - \omega_0 + \mu) \text{sech}\left(\beta \frac{\epsilon + \mu - \omega_0}{2}\right) \right] \left[\frac{\beta\mu}{2} \text{csch}\left(\frac{\beta\mu}{2}\right) \right]. \end{aligned} \quad (60)$$

Importantly, the expression above can be simplified in the limit $\beta\omega \rightarrow 0$. In this case, it is straightforward to check that $\sigma_{22}^c \propto T$.

In-plane resistivity with acoustic phonons

In this section, we consider the contribution to the in-plane resistivity due to the EME term enabled scattering off acoustic phonons. A well-defined (i.e., parametrically large) T^2 -scaling regime for $1/\tau_{\text{EME}}$ exists provided $J_H \ll \tilde{T}_{\text{BG}}$. We focus on temperatures, $J_H \ll T \ll \tilde{T}_{\text{BG}}$, with $\tilde{T}_{\text{BG}} = 2\tilde{c}k_F$ being the EME Bloch-Grüneisen (BG) temperature. In this regime, we only consider the effects of quasi-elastic scattering off the spins which have approximately local correlations, such that the spin structure factor can be approximated as a constant in momentum space *and* a delta function in frequency space [3]. In other words, phonons provide a source of inelastic scattering while the uncorrelated spins make this scattering isotropic. This is in contrast to the conventional electron-phonon interaction, where the scattering is mainly small-angle up to the BG temperature. A schematic for the scattering processes is shown in Fig. S3.



Supplemental Figure S3. A typical scattering event due to the EME interaction in the local approximation, where the spin structure factor is assumed to be featureless, i.e., $S(\omega, \mathbf{q}) \approx \text{const} \times \delta(\omega)$. Starting from $\mathbf{k}$ (center of green circle) on the Fermi surface (red circle), the local-moments scatter the momentum uniformly throughout the Brillouin zone via the EME term, such that the annulus (yellow) of phonon modes with momenta $\tilde{q}_{ph} \leq T/\tilde{c}$ efficiently scatters electrons isotropically on the Fermi surface. In contrast, the conventional el-ph interaction efficiently scatters electrons to a region enclosed by a sphere of radius $q_{ph} \approx T/c$, as shown in the green circle, such that scattering is mainly small-angle up to $\tilde{T}_{\text{BG}}$.

Let us first recall the contribution to the electron-phonon transport scattering time [8, 9] within Boltzmann theory (equivalent to Eq. (??)),

$$\frac{1}{\tau_{\text{el-ph}}} = \frac{4\pi}{\beta} \int_{\Omega} \frac{d\Omega}{\Omega} \alpha^2 F_{\text{tr}}(\Omega) \mathcal{K}(\beta\Omega). \quad (61)$$

In the Debye approximation for an acoustic mode in 2D, where $\alpha^2 F_{\text{tr}}(\Omega) \approx \lambda_{\text{el-ph}}(3/2) (\Omega/\Omega_D)^3 \theta(\Omega_D - \Omega)$, there is a factor of $(\Omega/\Omega_D)^2$ in the el-ph matrix element due to the weighting of small-angle scattering: $1 - \cos(\theta_{\mathbf{k}, \mathbf{k}'}) \sim q^2 \sim \Omega^2$, where $\mathbf{k}, \mathbf{k}'$ are points on the Fermi surface and $\mathbf{q}$ is the phonon wavevector, see e.g., [2]. In contrast, this weighting is absent in the isotropic EME transport time for reasons highlighted above, which can therefore be approximated as

$$\frac{1}{\tau_{\text{EME}}} = \frac{4\pi}{\beta} \int_{\Omega} \frac{d\Omega}{\Omega} \alpha^2 F_{\text{EME}}(\Omega) \mathcal{K}(\beta\Omega), \quad (62)$$

where $\alpha^2 F_{\text{EME}}(\Omega) \approx \frac{1}{2} \lambda_{\text{EME}} \left(\Omega/\tilde{\Omega}_{\text{BG}} \right) \theta(\tilde{\Omega}_{\text{BG}} - \Omega)$ (note that $\tilde{\Omega}_{\text{BG}} = \tilde{T}_{\text{BG}}$) and $\mathcal{K}$ defined by Eq. (30). Here we have assumed the BG temperature, rather than the Debye temperature, to be the relevant scale for transport. As described in Fig. S3, the dominant scattering is from thermally excited phonons, which can be seen here by the exponential suppression of $\mathcal{K}(x > 1)$. Scattering in this regime is therefore quasi-elastic. In total, we obtain that

$$\frac{1}{\tau_{\text{EME}}} \approx 2\pi \lambda_{\text{EME}} \frac{T^2}{\tilde{T}_{\text{BG}}}, \quad J_H \ll T \ll \tilde{T}_{\text{BG}}, \quad (63)$$

while for $T \gtrsim \tilde{T}_{\text{BG}}$, $1/\tau_{\text{EME}} \approx 2\pi\lambda_{\text{EME}}T$, as discussed earlier.

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